LLM ENGINEERING

LLM Evaluation Visual Guide

Proving a model works:
benchmarks, LLM judges, human eval, red teaming.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

Benchmarks

Standardized tests — with known limits.

Knowledge: MMLU

57 subjects, multiple choice — tests breadth of pretraining knowledge.

Reasoning: GSM8K / MATH

Grade-school to competition math — tests step-by-step reasoning.

Code: HumanEval

Write functions that pass hidden tests — tests practical coding.

Chat: MT-Bench / Arena

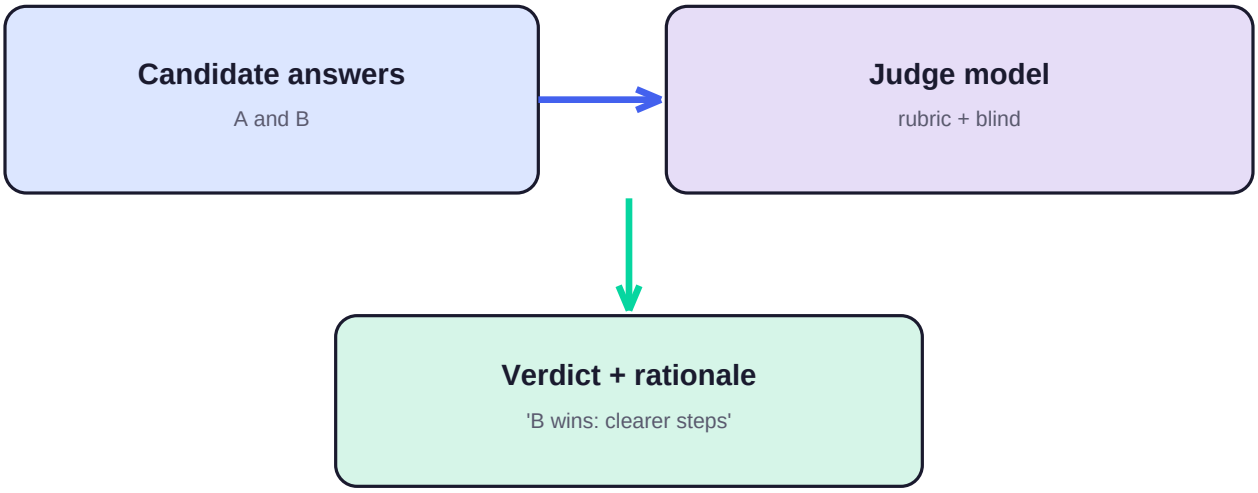
Multi-turn quality judged comparatively — closest to real use.

Benchmarks saturate and leak into training data. They rank models; they don't certify YOUR use case.

STEP 2/5

LLM-as-Judge

A strong model grades the weaker one.



Calibrate the judge against humans first. Watch for position bias (prefers answer A) and verbosity bias (prefers longer).

STEP 3/5

Human Evaluation

The gold standard — expensive and slow.

- Preference ranking: raters pick the better of two responses — simple, reliable, powers RLHF.
- Rubric scoring: rate helpfulness, accuracy, tone on 1-5 scales — more diagnostic.
- Inter-rater agreement (Cohen's kappa): low agreement means the task or rubric is underspecified.
- Blind the raters to which model produced what — branding bias is real.
- Use humans to calibrate automated evals, then scale with the automation.

STEP 4/5

Red Teaming

Adversarially probe for failures.

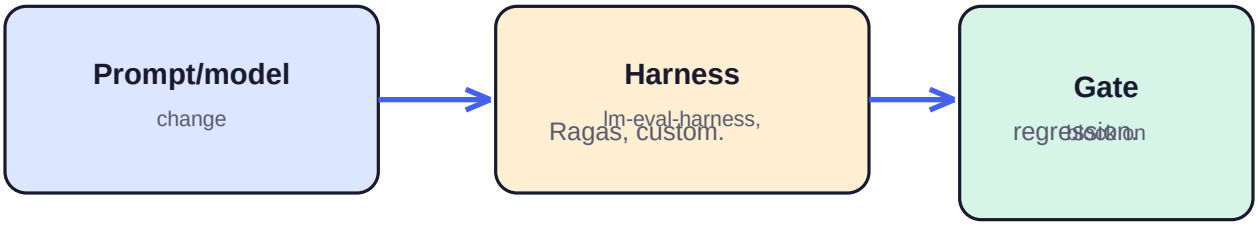
- Jailbreaks: prompt-injection and roleplay attacks that bypass safety behavior.
- Harmful outputs: disallowed content, biased decisions, dangerous instructions.
- Hallucination probes: obscure facts and edge cases where the model confidently invents.
- Track attack success rate per category; fix, then re-attack — red teaming is iterative.
- Include domain experts: generic testers miss failures only practitioners would catch.

Red team BEFORE launch, and after every major model or system-prompt change.

STEP 5/5

Eval Harnesses in CI

Evals as regression tests.



Pin eval versions and seeds. A flaky eval suite teaches the team to ignore it.

STEP ✓I✓

Key Takeaways

LLM Evaluation — Remember

- Benchmarks rank models but don't certify your use case; build task-specific evals.
- LLM-as-judge scales grading — calibrate against humans and correct for biases.
- Human eval is the gold standard: blind, rubric-driven, agreement-measured.
- Red team adversarially and iteratively, with domain experts included.
- Run evals in CI as regression gates on every prompt or model change.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev